

Introduction to Flutter

Easy Notes for Beginners and Non-IT Students (Simple English)

1. What is Flutter?

Imagine you want to make a new phone app. You want it to work on both **Android phones** and **iPhones (iOS)**.

- **The Old Way:** You must write the app two times. You write it once for Android using one language, and then write it again for iPhone using a different language. This takes a lot of time and money.
- **The Flutter Way:** You write the code **only one time**. Flutter takes that same code and makes it work on Android, iPhone, computers, and websites automatically.

So, **Flutter** is a free tool made by **Google**. It helps people build apps for many different screens using **only one set of code**.

Popular Apps Made with Flutter:

Google Pay Alibaba eBay Motors Google Ads NotebookLM

2. Native Apps vs. Cross-Platform Apps

These are just two different ways to make phone apps.

Feature	Native Apps (The Old Way)	Cross-Platform Apps (The Flutter Way)
Meaning	Made for only one type of phone.	Made once to work on many types of phones.
The Code	Separate code for Android and separate code for iPhone.	One single code shared by all devices.
Languages Used	Kotlin / Java (for Android) Swift (for iPhone)	Dart language (works for everything)

How the Work Looks

Without Flutter (Native Way)

Android App → Written in Java or Kotlin

iPhone App → Written in Swift

Web App → Written in JavaScript

Result: 3 different teams and 3 sets of code to look after.

With Flutter (New Way)

One Code → Written in Dart language

↓ Automatically gives you:

- Android App
- iPhone App
- Website
- Computer App

Result: 1 team and 1 set of code. Saves a lot of work!

3. The Good and the Bad

Both ways have things that are good and bad. It depends on what you need.

Why Native Apps are Good:

- **Good Performance:** They can run a little bit faster because they are built for only one phone.
- **Easy to Use Phone Parts:** They can easily connect to things inside the phone like the camera, GPS, or fingerprint scanner.

Why Flutter Apps are Good:

- **Faster to Make:** You do not need to write the same code many times.
- **Lower Cost:** Hiring one team to make one app is much cheaper than hiring two different teams.
- **Faster to Launch:** You can give your app to Android and iPhone users at the exact same time.

4. Why do People Choose Flutter?

- **Hot Reload (Instant Changes):** When a programmer changes a color or words in the code, they see the change on the phone screen instantly. They do not have to wait for the app to restart.
- **Ready-Made UI Blocks:** Flutter comes with pre-built pieces like buttons, text boxes, and menus. You can just grab them and use them like Lego blocks.
- **Better Design Control:** It lets you design the app exactly how you want it to look. The app will look the same on old phones and new phones.

5. Important: Flutter is NOT a Language

Many beginners make this mistake. Please remember this easy rule:

"Flutter is the Box of Tools. Dart is the Language."

Easy Example (Baking a Cake):

Think of making a cake in a kitchen:

- **Programming Language (Dart):** These are the raw food ingredients like flour, sugar, and eggs. You use them to write the basic instructions.
- **Framework (Flutter):** This is the kitchen setup. It gives you the electric mixer, the cake pans, and the oven. You do not make the tools yourself; you use them to bake the cake much faster.

Other Examples:

- **Languages (The Ingredients):** Dart, JavaScript, Python, Java
- **Frameworks (The Tools):** Flutter, React Native, Django